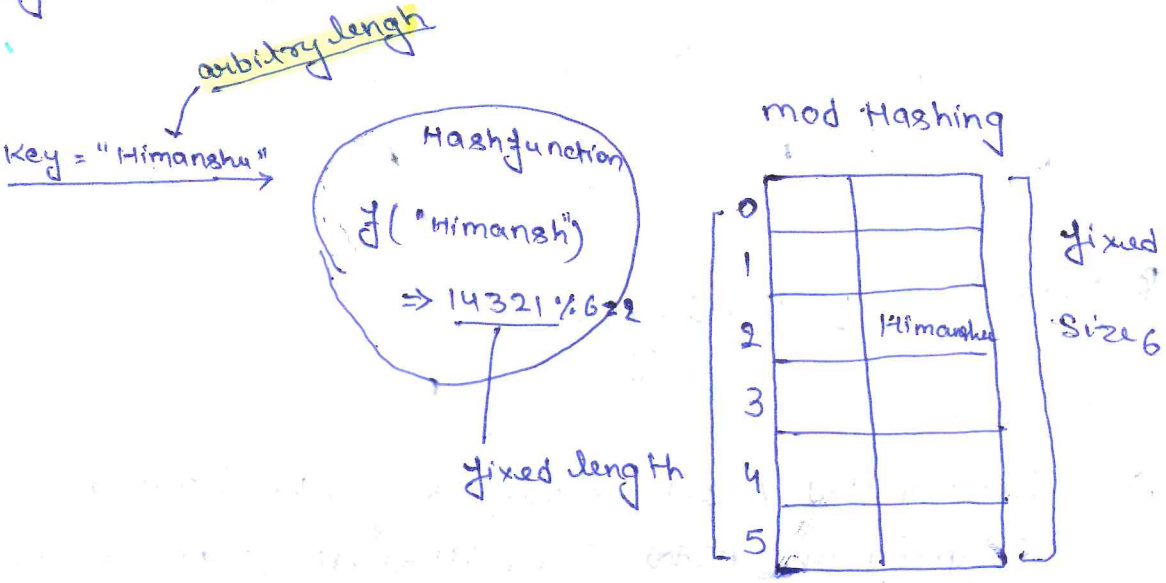
Consistent Hashing

Hashing :→



What is the Problem ??

If the Mod Hashing is fixed size that moment is No problem Otherwise size will be not fixed the problem occurs.

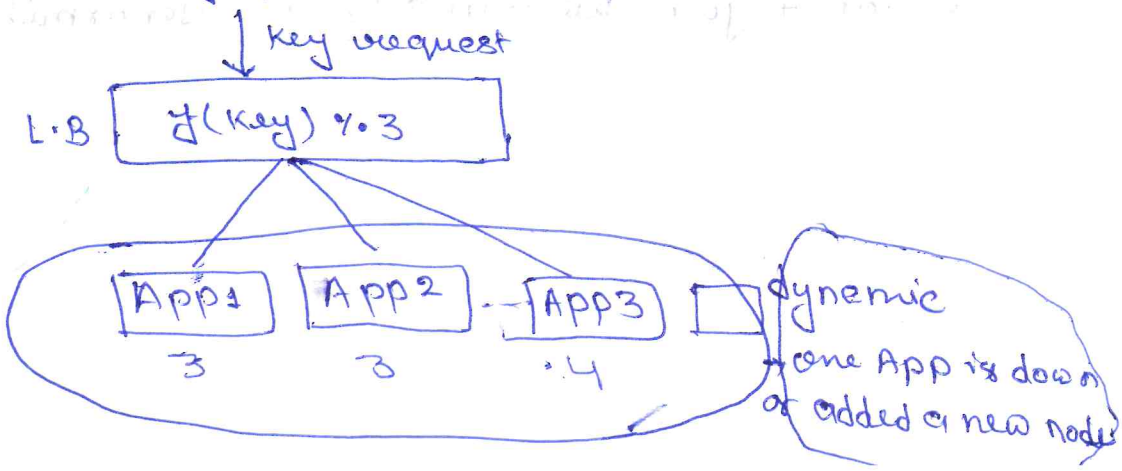
Concept && Coding

for example

① Load Balancing for App Server

② Horizontal Sharding

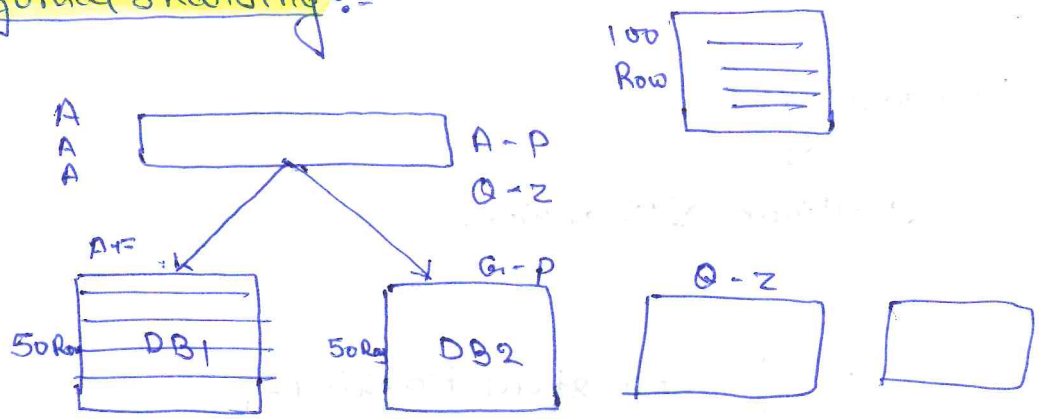
1: Load Balancing for Application Server :-



~~Note: LoadBalancer have distribute the re~~

Note:- LoadBalancer have equaly divide the traffic on my App Server. So that only one server don't have load. Suppose There are 10 request on my Server. So that LoadBalancer have devide equaly. There is not load on only One Server.

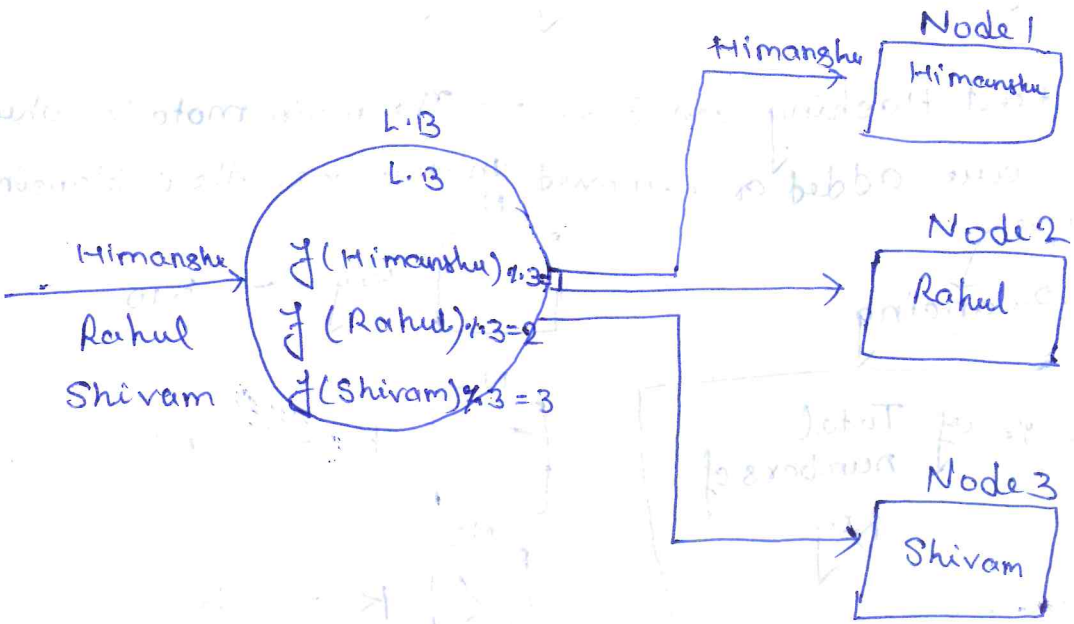
2: Horizontal Sharding:-



Concept && Coding

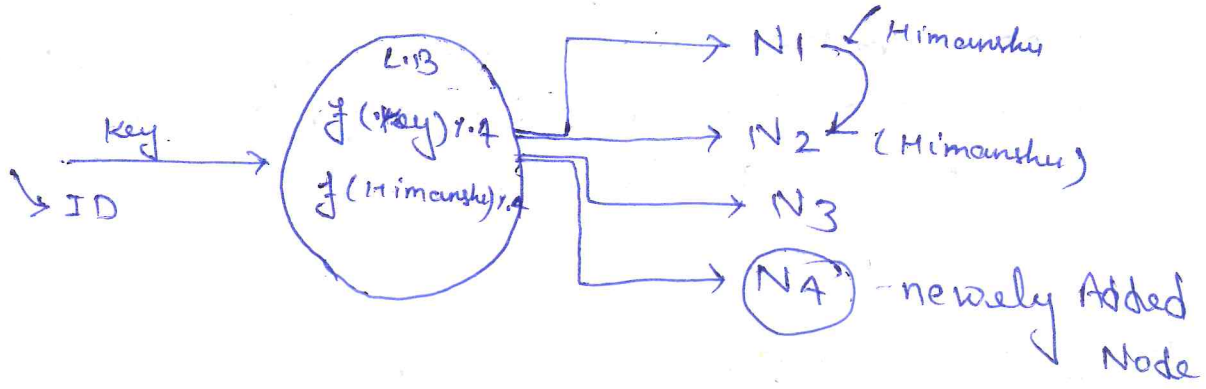
~~Suppose all name are A~~

Suppose more name with "A". that moment DB1 is early full. That time Load is not equaly Divided.



1 UseCase :-

Increase the Server or Node



Reblancing

→ Millions of DB entries

Consistent Hashing

Consistent hashing, rebalancing occurs when node (servers) are added or removed from the system. The goal is to minimize the number of keys that need to be reassigned while maintaing uniform distribution.

Concept && Coding

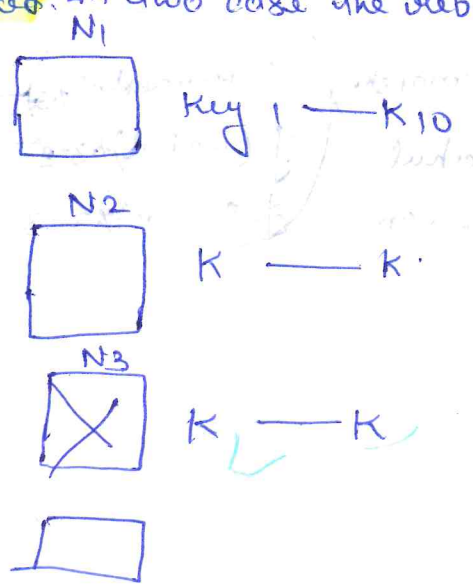
* Consistent Hashing can solve → The main moto is when node are added or removed. In two case the reblancing minimize

ReBalancing

$\frac{1}{N}$ % of Total numbers of Key

no of Node

good



How does Consistent Hashing Work??

① Virtual Ring

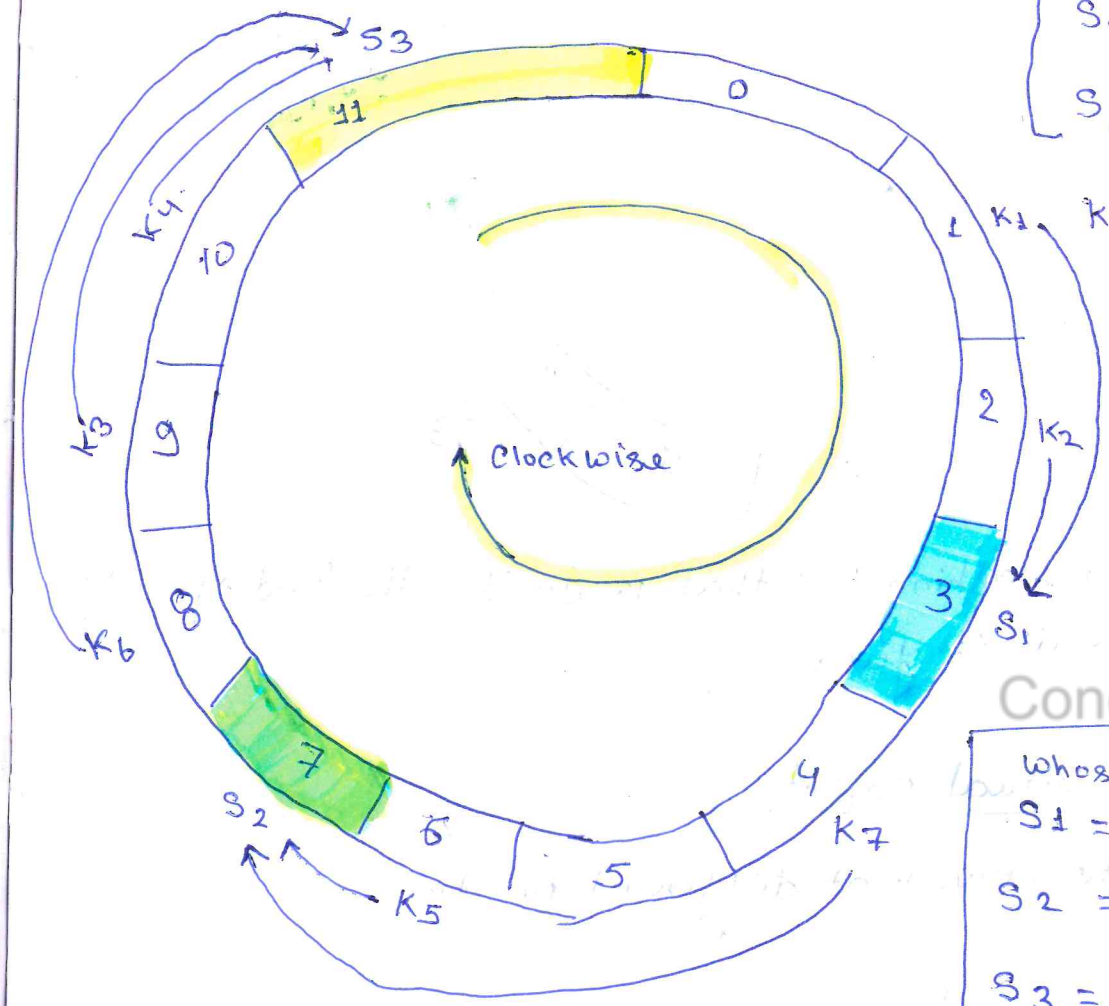
It brings

server = mod hash

S_1 = hash(Server1) % 12

S_2 = hash(Server2) % 12

S_3 = hash(Server3) % 12



key1 = has(key1) % 12

2

3

4

5

6

Concept && Coding

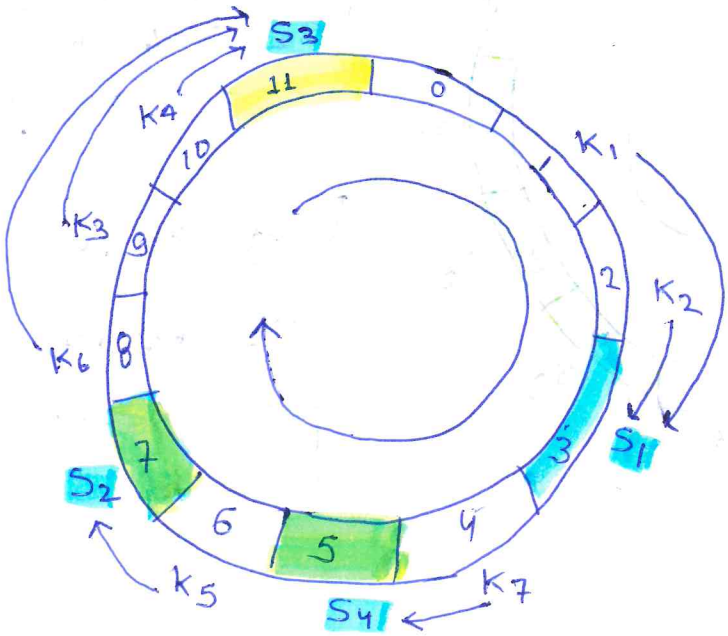
Whose server handle

S_1 = K_1, K_2

S_2 = K_5, K_7

S_3 = K_6, K_3, K_4

Addition of Servers



Whose server handle

S_1 = K_1, K_2

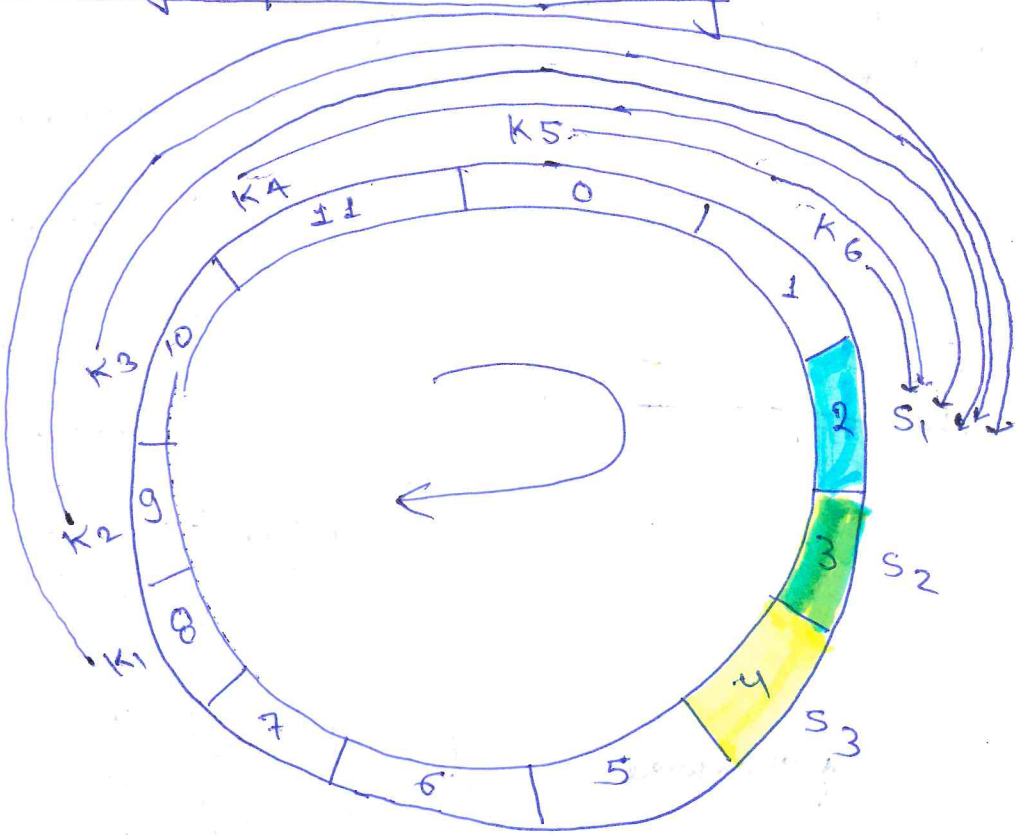
S_2 = K_5

S_3 = K_6, K_3, K_4

S_4 = K_7

* We can rebalance only Limited key

Disadvantages of Consistence Hashing :-

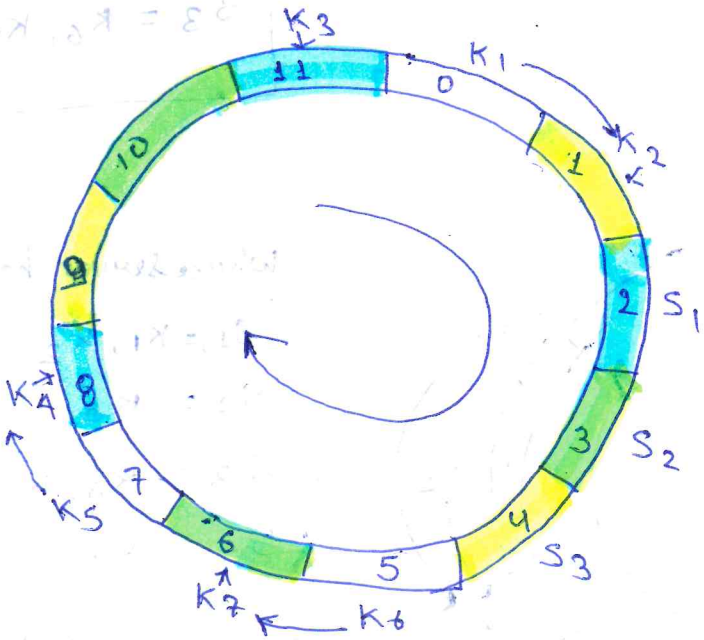


Consistence Hashing tell that is divide the load equally that time it is fail.

Concept && Coding

Create virtual Object

- replicate Server at different points.



Server replicas

$S_1 = 2, 8, 11$

$S_2 = 3, 6, 10$

$S_3 = 4, 9$

$S_3 = K_2, K_1$

$S_2 = K_6, K_7$

$S_1 = K_5, K_4, K_3$